

MATH 3260 - Chapter 1 Worksheets

Linear Systems

Name: _____

Date: _____

These are student practice worksheets. They are designed to be used with the chapter notebook and notes, but every problem below contains the data needed to work it independently. No solutions are included in this student version.

Section 1.1 - Systems of Linear Equations

Show enough work to make your reasoning clear. Interpret your answer when a problem asks for interpretation.

1. Decide whether each equation is linear in its variables.

(a) $3x - 4y + z = 21$

(b) $xy + z = 9$

(c) $x_1 - 2x_2 = 0$

(d) $\frac{1}{x} + y = 4$

2. Check whether $(4, 5)$ solves both equations

$$x + y = 9, \quad 2x - y = 3.$$

3. Classify the system as having one solution, no solution, or infinitely many solutions, and solve when appropriate:

$$x + y = 20, \quad x - y = 2.$$

4. Classify the system:

$$2x + 4y = 20, \quad x + 2y = 12.$$

Explain what algebraic contradiction or geometric relationship justifies your classification.

5. Write the complete solution of

$$x - y + 2z = 9$$

in parametric vector form.

6. Solve by elimination. Show the full system or augmented matrix after each elimination step:

$$\begin{cases} x + y + z = 13, \\ 2x + y - z = 13, \\ x - y + 2z = 1. \end{cases}$$

7. A fictional arena bundle contains two drinks and one snack for 21 dollars, while one drink and two snacks cost 24 dollars. Define variables, write a linear system, solve it, and interpret the modeled item prices.

Section 1.2 - Gaussian and Gauss-Jordan Elimination

Show enough work to make your reasoning clear. Interpret your answer when a problem asks for interpretation.

1. State the dimensions of

$$A = \begin{bmatrix} 1 & 9 & 20 \\ 21 & 4 & 7 \end{bmatrix}$$

and identify a_{23} .

2. Write the augmented matrix for

$$\begin{cases} x - 2y + z = 9, \\ 3x + y - 4z = 20. \end{cases}$$

3. Apply $R_2 \leftarrow R_2 - 3R_1$ to

$$\left[\begin{array}{cc|c} 1 & 2 & 9 \\ 3 & 7 & 30 \end{array} \right].$$

Show the entire matrix before and after the operation.

4. Classify each matrix as REF, RREF, or neither. Identify the leading entries or pivot 1s.

$$A = \begin{bmatrix} 1 & 2 & -1 & 4 \\ 0 & 3 & 5 & 7 \\ 0 & 0 & 2 & 6 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 2 & 5 \\ 0 & 1 & -3 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 3 & 0 \end{bmatrix}.$$

5. Solve by Gaussian elimination:

$$\begin{cases} x + y + z = 21, \\ 2x + y + 3z = 41, \\ x + 3y + 2z = 38. \end{cases}$$

6. Reduce to RREF and report the pivot columns using ordinary one-based column numbering:

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 20 \\ 1 & 2 & 1 & 42 \\ 2 & 1 & 1 & 40 \end{array} \right].$$

7. Classify and solve:

$$\begin{cases} x + 2y - z = 9, \\ 2x + 4y - 2z = 18. \end{cases}$$

8. Find all solutions of the homogeneous system

$$x + y + z = 0, \quad 2x + 2y + 2z = 0.$$

State the number of free variables.

Section 1.3 - Applications of Linear Systems

Show enough work to make your reasoning clear. Interpret your answer when a problem asks for interpretation.

1. Find the equation of the line through $(2, 21)$ and $(5, 9)$. Write the result in slope-intercept form.
2. Find the quadratic $p(x) = ax^2 + bx + c$ passing through $(0, 1)$, $(1, 9)$, and $(2, 21)$. Show the linear system for a, b, c before solving.
3. A node receives 50 units per hour from outside and sends flows x_1 and x_2 outward. Write the conservation equation.
4. Solve and interpret the one-parameter flow family

$$x_1 + x_2 = 50, \quad x_1 + 9 = x_3, \quad x_2 + x_3 = 59.$$

Let $x_1 = t$ and express all flows in terms of t .

5. Determine the feasible range of t in Problem 4 if every modeled flow must be nonnegative. Give one concrete feasible flow pattern.
6. Solve the circuit model

$$I_1 - I_2 + I_3 = 0, \quad 2I_1 + I_2 = 18, \quad I_2 + 3I_3 = 17.$$

Interpret the three numbers as branch currents and verify them in the original equations.

7. A three-channel calibration model satisfies

$$r + g + b = 21, \quad 2r + g + b = 30, \quad r + 2g + 3b = 44.$$

Find (r, g, b) and verify the result by substitution.